In [1]:
```
import numpy as np
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score,f1_score
from sklearn.model_selection import KFold,cross_val_score
```

In [2]:
```
iris=load_iris()
X=iris.data
y=iris.target
print("dataset shape:",X.shape)
```

```
dataset shape: (150, 4)
```

In [7]:
```
#Bootstrapping
np.random.seed(42)
n_iterations=10
bootstrap_accuracy=[]
bootstrap_f1=[]
for i in range(n_iterations):
    #generate bootstrap sample
    #sampling with replacement
    indices=np.random.choice(len(X),size=len(X),replace=True)

    X_train=X[indices]
    y_train=y[indices]


    test_indices=list(set(range(len(X)))-set(indices))

    #skip if no samples
    if(len(test_indices))==0:
       continue

    X_test=X[test_indices]
    y_test=y[test_indices]

    model=DecisionTreeClassifier(random_state=42)
    model.fit(X_train,y_train)

    y_pred=model.predict(X_test)

    accuracy=accuracy_score(y_test,y_pred)
    f1=f1_score(y_test,y_pred,average='weighted')

    bootstrap_accuracy.append(accuracy)
    bootstrap_f1.append(f1)

#Average bootstrap results
print("Bootstrapping Results")
print("---------------")
print("Average Accuracy:",
      round(np.mean(bootstrap_accuracy),3))
print("Average F1-Score:",
      round(np.mean(bootstrap_f1),3))
```

```
Bootstrapping Results
---------------
```

```
Average Accuracy: 0.937
Average F1-Score: 0.937
```

In [8]:

```
#K-Fold Cross Validation
kf=KFold(n_splits=5,shuffle=True,random_state=42)
model=DecisionTreeClassifier(random_state=42)

accuracy_scores=cross_val_score(model,X,y,cv=kf,scoring='accuracy')
f1_scores=cross_val_score(model,X,y,cv=kf,scoring='f1_weighted')

print("K-Fold Validation Results")
print("--------------------")
print("Average Accuracy:",
      round(np.mean(accuracy_scores),3))
print("Average F1-Score:",
      round(np.mean(f1_scores),3))
```

```
K-Fold Validation Results
--------------------
Average Accuracy: 0.953
Average F1-Score: 0.953
```